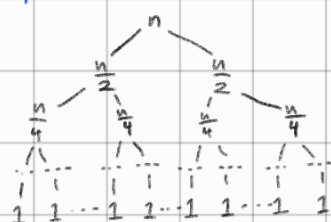


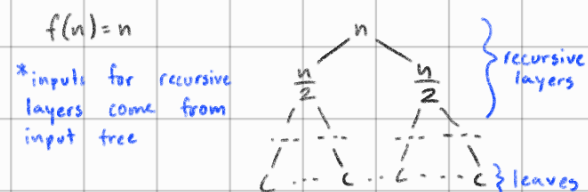
Recursion Trees

ex. given $s(1) = c$, $s(n) = 2s(\frac{n}{2}) + n$ $n \geq 2$, draw recursion tree

① draw input-size tree



② recursion tree (depends on non-recursive work in recursive formula)



③ Calculate total internal work

At level 0, 1 node w/ n

level 1, 2 nodes w/ $\frac{n}{2}$

level 2, 4 nodes w/ $\frac{n}{4}$

At each internal level, total work equals n ($1 \cdot n = 2 \cdot \frac{n}{2} = 4 \cdot \frac{n}{4} = \dots$)

How many internal levels?

$n \rightarrow \frac{n}{2} \rightarrow \frac{n}{4} \rightarrow \dots \rightarrow 1$
 $\log_2 n$ times, so $\log_2 n$ levels

Each level is n , $\log n$ levels, so total internal work is $n \log n$

④ Calculate work from leaves

Two children per node, $\log_2 n$ levels, so $2^{\log_2 n} = n$ leaves

Each leaf has time c so total work for leaves is cn

⑤ Calculate total work

Total work for entire tree is $n \log n + cn$

Heuristic for finding number of internal levels:

$$T(n) = c \cdot T\left(\frac{n}{x}\right) + \dots$$

↳ each time you divide input size by x , so there are $\log_x n$ layers

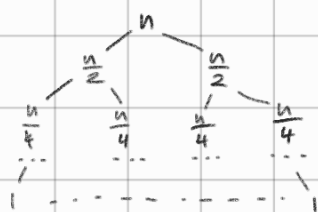
$$T(n) = aT(n-b) + \dots \quad T(c) = d$$

total levels is $\frac{n-c}{b} + d$

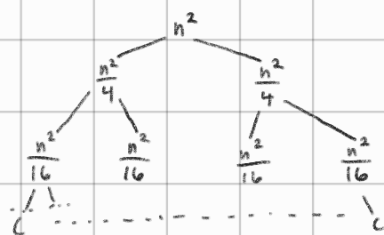
*This step is wrong in the lecture, if there are $\log_2 n$ total levels, there should be $\log_2 n - 1$ internal levels. This step is done correctly in the next example

ex. $P(1) = c$ $P(n) = 2P(\frac{n}{2}) + n^2$ $n \geq 2$

①



② $g(n) = n^2$



③

level 0: n^2
 1: $(\frac{n}{2})^2 \cdot 2$
 2: $(\frac{n}{4})^2 \cdot 4$
 k: $(\frac{n}{2^k})^2 \cdot 2^k$

sum:

$$\sum_{i=0}^k \left(\frac{n}{2^i}\right)^2 \cdot 2^i = \sum_{i=0}^k \frac{n^2}{2^i} = n^2 \sum_{i=0}^k \frac{1}{2^i}$$

Takes $\log_2 n$ levels to reach base case (see heuristic), $k = \log_2 n - 1$

$$n^2 \sum_{i=0}^{\log_2 n - 1} \frac{1}{2^i} = n^2 \left(2 - \frac{1}{2^{\log_2 n - 1}}\right) = n^2 \left(2 - \frac{2}{n}\right) = 2n^2 - 2n$$

④

$2^{\log_2 n} = n$ leaves, total leaf work is cn

⑤

total work for entire tree is $2n^2 - 2n + cn$, so $O(n^2)$

General Heuristic for leaves:

$$T(n) = aT\left(\frac{n}{b}\right) \rightarrow \text{leaves} = a^{\log_b n}$$

$$\hookrightarrow \text{levels} = \log_b n$$

13.1 Recursion trees

Use recursion trees to find closed forms for the following. Ensure that the input is always an integer by assuming, as needed, that n has a special form (e.g. a power of 2).

- (a) $T(1) = 47$
 $T(n) = 3T(n/3) + 13n$ for $n \geq 2$
- (b) $T(1) = 1$
 $T(n) = 2T(n-1) + 3$ for $n \geq 2$

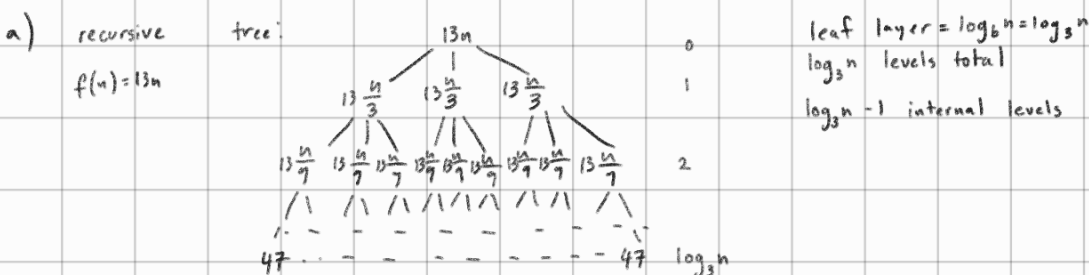


table:

level	problem size	nodes	work/node	level total
0	n	1	$13n$	$13n$
1	$\frac{n}{3}$	3	$13(\frac{n}{3})$	$13n$
2	$\frac{n}{3^2}$	3^2	$13(\frac{n}{3^2})$	$13n$
3	$\frac{n}{3^3}$	3^3	$13(\frac{n}{3^3})$	$13n$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
h	$\frac{n}{3^h} = 1$	3^h	$T(1) = 47$	$47 \cdot 3^h \leftarrow \text{leaves}$

$h = \log_3 n$

$$3^h = 3^{\log_3 n} = n \text{ leaves, } \therefore \text{total leaf work} = 47n$$

$$\text{non-leaf work (internal)} = \sum_{k=0}^{\log_3 n - 1} 13n = 13n \log_3 n$$

$$\text{total work} = 47n + 13n \log_3 n$$

- b) $T(1) = 1$ $T(n) = 2T(n-1) + 3$ for $n \geq 2$

$$\frac{n-c}{b} + d \quad f(n) = 3$$

$$\frac{n-1}{1} + 1 = n$$

n total levels

leaves at $n-1$

internals end at $n-2$

$$\text{leaf work} = 2^{n-1}$$

$$\text{internal work} = 3 \sum_{i=0}^{n-2} 2^i = (2^{n-1} - 1)3 = 3 \cdot 2^{n-1} - 3$$

$$\text{total work} = 2^{n-1} + 3 \cdot 2^{n-1} - 3 = 2^{n+1} - 3$$

level	prob. size	nodes	work/node	level total
0	n	1	3	3
1	$n-1$	2	3	$3 \cdot 2$
2	$n-2$	2^2	3	$3 \cdot 2^2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$n-k$	2^k	3	$3 \cdot 2^k$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$n-1$	$n-(n-1)=1$	2^{n-1}	$T(1)=1$	$1 \cdot 2^{n-1} \leftarrow \text{leaf level}$

